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Documentation

(<https://docs.aws.amazon.com/index.html>)



Amazon DynamoDB

(<https://docs.aws.amazon.com/dynamodb/index.html>)



D...

What is Amazon DynamoDB?

[↓ PDF \(/pdfs/amazondynamodb/latest/developer%2Fintroduction.html\)](#)

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- [Getting started\(#ddb-intro-get-started\)](#)

Related resources

- [Amazon DynamoDB API Reference \(https://docs.aws.amazon.com/amazondynamodb/latest/APIReference/index.html\)](https://docs.aws.amazon.com/amazondynamodb/latest/APIReference/index.html)
- [Cheat Sheet \(https://docs.aws.amazon.com/amazondynamodb/latest/developer%2Fintroduction.html\)](https://docs.aws.amazon.com/amazondynamodb/latest/developer%2Fintroduction.html)
- [Using Amazon DynamoDB with an AWS SDK \(https://docs.aws.amazon.com/amazondynamodb/latest/developer%2Fintroduction.html\)](https://docs.aws.amazon.com/amazondynamodb/latest/developer%2Fintroduction.html)

▼ Recommended tasks

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(<https://docs.aws.amazon.com/lambda/latest/dg/welcome.html>)

[Understand the features and capabilities of Amazon S3](https://docs.aws.amazon.com/AmazonS3/latest/userguide/Welcome.html)

(<https://docs.aws.amazon.com/AmazonS3/latest/userguide/Welcome.html>)

[Understand core components and data organization in DynamoDB](https://docs.aws.amazon.com/amazondynamodb/latest/developerguide/HowItWorks.CoreComponents.html)

(<https://docs.aws.amazon.com/amazondynamodb/latest/developerguide/HowItWorks.CoreComponents.html>)

[Understand what Amazon Relational Database Service is](https://docs.aws.amazon.com/AmazonRDS/latest/UserGuide/Welcome.html)

(<https://docs.aws.amazon.com/AmazonRDS/latest/UserGuide/Welcome.html>)

[Understand the features and capabilities of Amazon Aurora](https://docs.aws.amazon.com/AmazonRDS/latest/AuroraUserGuide/CHAP_AuroraOverview.html)

(https://docs.aws.amazon.com/AmazonRDS/latest/AuroraUserGuide/CHAP_AuroraOverview.html)

How to

[Create and manage DynamoDB tables for database development](https://docs.aws.amazon.com/amazondynamodb/latest/developerguide/GettingStartedDynamoDB.html)

(<https://docs.aws.amazon.com/amazondynamodb/latest/developerguide/GettingStartedDynamoDB.html>)

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Amazon DynamoDB is a serverless, fully managed, distributed NoSQL database with single-digit millisecond performance at scale.

DynamoDB addresses your needs to overcome scaling and operational complexities of relational databases. It is purpose-built and optimized for operational workloads that require consistent performance. DynamoDB delivers consistent single-digit millisecond performance for a shopping cart use case with 10 million users. [Launched in 2012](https://press.aboutamazon.com/2012/1/amazon-web-services-nosql-database-service-designed-for-the-scale-of-the-internet) [↗](https://press.aboutamazon.com/2012/1/amazon-web-services-nosql-database-service-designed-for-the-scale-of-the-internet) (<https://press.aboutamazon.com/2012/1/amazon-web-services-nosql-database-service-designed-for-the-scale-of-the-internet>) , DynamoDB continues to help you build scalable applications and databases while reducing cost and improving performance at scale.

Customers across all sizes, industries, and geographies use DynamoDB to build modern, serverless applications that scale globally. DynamoDB scales to support tables of virtually any size while providing millisecond performance and high availability.

For events, such as [Amazon Prime Day](https://aws.amazon.com/blogs/aws/prime-day-2023-peak-traffic/) [↗] (<https://aws.amazon.com/blogs/aws/prime-day-2023-peak-traffic/>), DynamoDB powers multiple high-traffic Amazon properties and systems, including [Alexa](https://www.amazon.com/) [↗] (<https://www.amazon.com/>) sites, and all [Amazon fulfillment centers](https://www.aboutamazon.com/news/infrastructure/amazon-fulfillment-centers) [↗] (<https://www.aboutamazon.com/news/infrastructure/amazon-fulfillment-centers>). For events, DynamoDB APIs have handled trillions of calls from Amazon properties and systems. For hundreds of customers with tables that have peak traffic of over half a million requests per second, DynamoDB has supported customers whose table sizes exceed 200 TB, and processes over one billion requests per hour.

Topics

- [Characteristics of DynamoDB](#) (#ddb-characteristics)
- [DynamoDB use cases](#) (#ddb-use-cases)
- [Capabilities of DynamoDB](#) (#ddb-capabilities)
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- [Accessing DynamoDB](#) (#ddb-access)
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- [Getting started with DynamoDB](#) (#ddb-intro-get-started)

Characteristics of DynamoDB

Serverless

With DynamoDB, you don't need to provision any servers, or patch, manage, install, maintain, or upgrade software. DynamoDB provides zero downtime maintenance. It has no versions (major, minor, or patch), or release windows.

DynamoDB's [on-demand capacity mode](#) ([./on-demand-capacity-mode.html](#)) offers pay-as-you-go pricing for requests so you only pay for what you use. With on-demand, DynamoDB instantly scales up to handle any traffic and maintains performance with zero administration. It also scales down to zero so your table doesn't have traffic and there are no cold starts.

NoSQL

As a NoSQL database, DynamoDB is purpose-built to deliver improved performance, scalability compared to traditional relational databases. To support a wide variety of use cases, DynamoDB supports document data models.

Unlike relational databases, DynamoDB doesn't support a JOIN operator. We recommend that you reduce database round trips and processing power needed to answer queries. As a NoSQL database, DynamoDB offers strong [read consistency](#) ([./HowItWorks.ReadConsistency.html](#)) and [ACID transactions](#) [L](https://aws.amazon.com/dynamodb-transactions/) (<https://aws.amazon.com/dynamodb-transactions/>) to build enterprise-grade applications.

Fully managed

As a fully managed database service, DynamoDB handles the undifferentiated heavy lifting so you can focus on building value for your customers. It handles setup, configurations, maintenance, provisioning, security, backups, monitoring, and more. This ensures that when you create a database for production workloads, DynamoDB constantly improves its availability, reliability, performance without requiring upgrades or downtime.

Single-digit millisecond performance at any scale

DynamoDB was purpose-built to improve upon the performance and scalability of relational databases, delivering single-digit millisecond performance at any scale. To achieve this scale and performance, DynamoDB is optimized for a wide range of workloads and provides APIs that encourage efficient database usage. It omits features that are not needed at scale, for example, JOIN operations. DynamoDB delivers consistent single-digit millisecond performance whether you have 100 or 100 million users.

DynamoDB use cases

Customers across all sizes, industries, and geographies use DynamoDB to build modern, serverless applications that scale small and scale globally. DynamoDB is ideal for use cases that require consistent performance and low operational overhead. The following list presents some use cases where you can use DynamoDB.

- **Financial service applications** – Suppose you're a financial services company building applications for routing, loan management, token generation, and transaction ledgers. With DynamoDB, your applications can respond to events and serve traffic from your chosen AWS Regions with low latency and high performance.

DynamoDB is suitable for applications with the most stringent availability requirements. You can scale instances of manually scaling instances for increased storage or throughput, versioning, and licensing. You can use [DynamoDB transactions](#) ([./transactions.html](#)) to achieve atomicity, consistency, and isolation across one or more tables with a single request. [\(ACID\) transactions](#) ([./transaction-apis.html](#))

processing financial transactions or fulfilling orders. DynamoDB instantly accommodates down, enabling you to efficiently scale your database for market conditions, such as trad

- **Gaming applications** – As a gaming company, you can use DynamoDB for all parts of ga state, player data, session history, and leaderboards. Choose DynamoDB for its scale, con of operations provided by its serverless architecture. DynamoDB is well suited for scale-c successful games. It quickly scales your game's throughput both in and out (scale to zero optimizes your architecture's efficiency whether you're scaling out for peak traffic or scal low.
- **Streaming applications** – Media and entertainment companies use DynamoDB as a meti management service, or to serve near real-time sports statistics. They also use DynamoD bookmarking services and process billions of daily customer events for generating recor benefit from DynamoDB's scalability, performance, and resiliency. DynamoDB scales to w or down, enabling streaming media use cases that can support any levels of demand.

To learn more about how customers from different industries use DynamoDB, see [Amazon D](https://aws.amazon.com/dynamodb/customers/) (<https://aws.amazon.com/dynamodb/customers/>) and [This is My Architecture](https://aws.amazon.com/architecture/?tma.sort-by=item.additionalFields.airDate&tma.sort-order=desc&awsf.category=*all&awsf.industry=*all&awsf.language=*all&awsf.show=*all&awsf.products=AND) [L](https://aws.amazon.com/architecture/?tma.sort-by=item.additionalFields.airDate&tma.sort-order=desc&awsf.category=*all&awsf.industry=*all&awsf.language=*all&awsf.show=*all&awsf.products=AND) (https://aws.amazon.com/architecture/?tma.sort-by=item.additionalFields.airDate&tma.sort-order=desc&awsf.category=*all&awsf.industry=*all&awsf.language=*all&awsf.show=*all&awsf.products=AND) .

Capabilities of DynamoDB

Multi-active replication with global tables

[Global tables](#) ([./GlobalTables.html](#)) provide multi-active replication of your data across your ch availability [L](https://aws.amazon.com/dynamodb/sla/) (<https://aws.amazon.com/dynamodb/sla/>) . Global tables deliver a fully managed s Region, multi-active database, without building and maintaining your own replication solutio specify the AWS Regions where you want the tables to be available. DynamoDB replicates on tables.

Your globally distributed applications can access data locally in your selected Regions to achi and write performance. Because global tables are multi-active, you don't need a primary tab complicated or delayed fail-overs, or database downtime when failing over an application be

ACID transactions

DynamoDB is built for mission-critical workloads. It includes [\(ACID\) transactions](#) ([./transaction](#) that require complex business logic. DynamoDB provides native, server-side support for tran experience of making coordinated, all-or-nothing changes to multiple items within and across

Change data capture for event-driven architectures

DynamoDB supports streaming of item-level change data capture (CDC) records in near-real time models for CDC: [DynamoDB Streams](#) ([./Streams.html](#)) and [Kinesis Data Streams for DynamoDB](#). When an application creates, updates, or deletes items in a table, streams records a time-ordered sequence of changes in near-real time. This makes DynamoDB Streams ideal for applications with event-driven architectures.

Secondary indexes

DynamoDB offers the option to create both [global and local secondary indexes](#) ([./SecondaryIndexes.html](#)) on table data using an alternate key. With these secondary indexes, you can access data with attributes that are not in the primary key, giving you maximum flexibility in accessing your data.

Service integrations

DynamoDB broadly integrates with several AWS services to help you get more value from your data, reduce operational overhead, and operate your workloads at scale. Some examples are: AWS CloudFormation, AWS IAM, AWS Identity and Access Management (IAM), and AWS Auto Scaling. The following sections cover integrations that you can perform using DynamoDB:

Serverless integrations

To build end-to-end serverless applications, DynamoDB integrates natively with a number of AWS services. For example, you can integrate DynamoDB with AWS Lambda to [create triggers](#) ([./Streams.LambdaTriggers.html](#)) that automatically respond to events in DynamoDB Streams. With triggers, you can build event-driven applications that respond to data modifications in DynamoDB tables. For cost optimization, you can [filter events](#) ([./Streams.Filtering.html](#)) to remove unwanted processes from a DynamoDB stream.

The following list presents some examples of serverless integrations with DynamoDB:

- [AWS AppSync](#) (<https://docs.aws.amazon.com/appsync/latest/devguide/what-is-appsync.html>) for GraphQL
- [Amazon API Gateway](#) (<https://docs.aws.amazon.com/apigateway/latest/developerguide/welcome.html>) for REST APIs
- [Lambda](#) (<https://docs.aws.amazon.com/lambda/latest/dg/welcome.html>) for serverless computation
- [Amazon Kinesis Data Streams](#) (<https://docs.aws.amazon.com/streams/latest/dev/introduction.html>) for real-time data streaming

Importing and exporting data to Amazon S3

Zero-ETL integration

Caching

Security

By default, DynamoDB encrypts all customer data at rest. [Encryption at rest \(./EncryptionAtRest\)](#) your data by using encryption keys stored in [AWS Key Management Service](#) (<https://docs.aws.amazon.com/kms/latest/developerguide/overview.html>) (AWS KMS). With encryption, you can protect sensitive applications that meet strict encryption compliance and regulatory requirements. V DynamoDB decrypts the table data transparently. You don't have to change any code or appl

encrypted tables. DynamoDB continues to deliver the same single-digit millisecond latency to all [DynamoDB queries](#) ([./Query.html](#)) work seamlessly on your encrypted data.

You can specify whether DynamoDB should use an AWS owned key (default encryption type) managed key to encrypt user data. The default encryption using [AWS-owned KMS keys](#) (<https://docs.aws.amazon.com/kms/latest/developerguide/concepts.html#key-mgmt>) is available at encryption, you can use the [AWS Database Encryption SDK](#) [↗](#) (<https://aws.amazon.com/blogs/encryption-sdk-for-client-side-encryption-and-perform-searches-on-encrypted-attributes-in-dynamodb/>)

DynamoDB also adheres to several [compliance standards](#) [↗](#) (<https://aws.amazon.com/compliance/>) including HIPAA, PCI DSS, and GDPR, which enables you to meet regulatory requirements.

Resilience

By default, DynamoDB automatically replicates your data across three [Availability Zones](#) [↗](#) (https://aws.amazon.com/global-infrastructure/regions_az/) to provide high durability and a 99.99% availability SLA. capabilities to help you achieve your business continuity and disaster recovery objectives.

DynamoDB includes the following features to help support your data resiliency and backup r

Features

- [Global tables](#) ([#ddb-resilience-gt](#))
- [Continuous backups and point-in-time recovery](#) ([#ddb-resilience-backups-pitr](#))
- [On-demand backup and restore](#) ([#ddb-resilience-ondemand-backup-restore](#))

Global tables

DynamoDB global tables enable a [99.999% availability SLA](#) [↗](#) (<https://aws.amazon.com/dynamodb/global-tables/>) to provide high durability and resilience. This helps you build resilient applications and optimize them for the lowest recovery point objective (RPO). Global tables also integrates with [AWS Fault Injection Service](#) (<https://docs.aws.amazon.com/fis/latest/userguide/what-is.html>) to perform fault injection experiments on your workloads. For example, [pausing global table replication](#) (<https://docs.aws.amazon.com/fis/latest/userguide/what-is.html#dynamodb-actions-reference>) to any replica table.

Continuous backups and point-in-time recovery

[Continuous backups](#) ([./Point-in-time-recovery.html](#)) provide you per-second granularity and the ability to restore a table to any point in time up to the last 35 days.

can set the recovery period to any value between 1 and 35 days.

Continuous backups and initiating a point-in-time restore doesn't use provisioned capacity. This helps maintain the performance or availability of your applications.

On-demand backup and restore

[On-demand backup and restore \(./Backup-and-Restore.html\)](#) let you create full backups of a table and store them in an Amazon S3 bucket. Backups don't impact the performance of your table regardless of any size. With [AWS Backup integration \(./Backup-and-Restore.html\)](#), you can use AWS Backup to manage the life cycle of your DynamoDB on-demand backups automatically. Using AWS Backup, you can manage backups across accounts and Regions, and transition older backups to cold storage for cost-optimization.

Accessing DynamoDB

You can work with DynamoDB using the [AWS Management Console](#) [\(https://console.aws.amazon.com/dynamodb/\)](https://console.aws.amazon.com/dynamodb/), [Command Line Interface](#) [\(https://aws.amazon.com/cli/\)](https://aws.amazon.com/cli/), [NoSQL Workbench for DynamoDB](#) [\(./NoSQL-Workbench-for-DynamoDB.html\)](#), or [DynamoDB APIs](#) [\(https://docs.aws.amazon.com/amazondynamodb/latest/APIReference/API_Operations_AmazonDynamoDB.html\)](https://docs.aws.amazon.com/amazondynamodb/latest/APIReference/API_Operations_AmazonDynamoDB.html).

For more information, see [Accessing DynamoDB \(./AccessingDynamoDB.html\)](#).

DynamoDB pricing

DynamoDB charges for reading, writing, and storing data in your tables, along with any optional features. DynamoDB has two capacity modes with their respective billing options for processing reads and writes: [on-demand \(./on-demand-capacity-mode.html\)](#) and [provisioned \(./provisioned-capacity-mode.html\)](#).

DynamoDB is also included in the **always free tier**, providing 25 GB of storage. The **Always Free** tier also includes 1000 Write and 25 provisioned Read Capacity Units (WCU, RCU) which is enough to handle 200 MB of data.

For more information, see [Amazon DynamoDB pricing](#) [\(https://aws.amazon.com/dynamodb/pricing/\)](https://aws.amazon.com/dynamodb/pricing/).

Getting started with DynamoDB

If you're a first-time user of DynamoDB, we recommend that you begin by reading the following guides:

- [Getting started with DynamoDB \(./GettingStartedDynamoDB.html\)](#) – Walks you through the creating sample tables, and uploading data. This topic also provides information about p operations using the AWS Management Console, AWS CLI, NoSQL Workbench, and Dyna
- [DynamoDB core components \(./HowItWorks.CoreComponents.html\)](#) – Describes the basic Dy
- [Best practices for designing and architecting with DynamoDB \(./best-practices.html\)](#) – Prov NoSQL design, DynamoDB Well-Architected Lens, table design and several other Dynam help you maximize performance and minimize throughput costs when working with Dyn

We also recommend that you review the following tutorials that present complete end-to-er with DynamoDB. You can complete these tutorials using the **always free tier** feature.

- [Create and Query a NoSQL Table with Amazon DynamoDB](#)  (<https://aws.amazon.com/tuto>)
- [Build an Application Using a NoSQL Key-Value Data Store](#)  (<https://aws.amazon.com/tuto>
[sql-key-value-data-store/](#))

For information about resources, tools, and strategies to migrate to DynamoDB, see [Migratin guide.html#migration-guide.title](#) . To read the latest blogs and whitepapers, see [Amazon Dynar](#) (<https://aws.amazon.com/dynamodb/resources/>) .

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DynamoDB core components: tables, items, attributes, primary keys,

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[Using Global Secondary Indexes in DynamoDB...](#)

DynamoDB global secondary indexes enable efficient queries on non-key attributes, supporting leaderboard

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